

Connectivity and Consumerisation Drive Renewables



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On October 2, 2015, at Times Square in New York City, USA, the renewable energy sector's most efficient solar panel was announced by SolarCity. Clocked at 22.04 per cent efficiency by Renewable Energy Test Centre, these cells enable users to extract maximum power from the large fusion explosion in the sky that we endearingly call our Sun. What other technology advances drive the renewable sector?

Technology updates affecting renewables

Countries like Switzerland are already working on restructuring their entire energy supply systems. The new supply mix will be free from nuclear power, low in carbon intensity and resting upon much higher efficiencies based on the newest and the most energy-efficient technologies. The new systems will also rely on development of smart-grids, decentralised power suppliers, hydro power, wind power, photovoltaics, biomass, wood and rigorous use of burning waste to generate energy whenever materials cannot be recycled. Switzerland most likely has to find its own energy supply mix, with the biggest sustainability potential. How will they do it?

Bernard Bonin and colleagues from Atomic Energy Research Centre CEA Saclay, France, simulated the behaviour of the energy mix for a large number of such vari-

ables using so-called Monte-Carlo simulations. They found the energy cost of the mix presents a minimum as a function of the installed power. This means that if installed power is too large, fixed costs dominate the total and become overwhelming. In contrast, if it is too small, expensive energy sources need to be frequently solicited.

Ted Brekken, an associate professor and renewable energy expert at College of Engineering, Oregon State University, USA, has done an analysis that suggests that large-scale wave-energy systems developed in the Pacific-Northwest are comparatively steady, dependable and able to be integrated into the overall energy grid at lower costs than some other forms of alternative energy.

The single-most significant advancement in technology is real-time monitoring of all aspects of our energy-generation and delivery system. "We must eliminate the waste that exists in the open-loop systems deployed globally. This is the only way we will reduce the impact on the environment, improve efficiency of energy extraction and delivery, and provide our children with a clean and sustainable energy future using solutions that integrate all elements of real-time monitoring, including machine-to-machine (M2M) learning that builds intelligence into the monitoring system," says Lyle Shuey, consulting vice president - Americas, Product Engineering Services, Embitel Technologies.